

Mathematics Extension 2

Trial Higher School Certificate Examination

2010

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Total Marks – 120

Attempt Questions 1-8

All Questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Begin a NEW writing booklet			Marks
a)	Evaluate	$\int_{\frac{\pi}{16}}^{\frac{\pi}{12}} \sec 4x \tan 4x \, dx$	2
b)	Find	$\int x \ln x \, dx$	2
c)	Find	$\int \frac{9x^3 + 9x^2 + 5x + 4}{3x + 1} \, dx$	3
d)	i.	Find constants a, b and c such that	2
		$\frac{3x^2 - 2x - 3}{x^2 + 9} \cdot \frac{1}{x - 3} = \frac{ax + b}{x^2 + 9} + \frac{c}{x - 3}$	
	ii.	Hence find $\int \frac{3x^2 - 2x - 3}{x^2 + 9} \cdot \frac{1}{x - 3} \, dx$.	2
e)	By making the substitution $t = \tan \frac{\theta}{2}$, evaluate	$\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta + \cos \theta}$	4

End of Question 1

Question 2 (15 Marks)

Begin a NEW booklet

Marks

- a) Given $A = 3 + 4i$ and $B = 1 - i$, express the following in the form $x + iy$ where x and y are real numbers.

i. AB

1

ii. $\frac{A}{iB}$

2

iii. $\sqrt{A}$

3

- b) If $w = \sqrt{3} - i$,

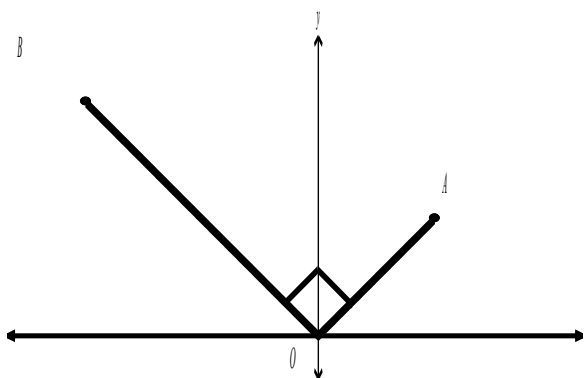
i. Find the exact value of $|w|$ and $\arg w$.

2

ii. Find the exact value of w^5 in the form $a + ib$ where a and b are real.

2

- c)



On the Argand diagram, OA represents the complex number $z_1 = x + iy$, $\angle AOB = \frac{\pi}{2}$ and the length of OB is twice that of OA .

i. Show that OB represents the complex number $-2y + 2ix$.

1

ii. Given that $AOBC$ is a rectangle, find the complex number represented by OC .

1

iii. Find the complex number represented by BA .

1

- d) Sketch the region on an argand diagram where

2

$$|z - 1| \geq \sqrt{2} \text{ and } 0 \leq \arg(z + i) \leq \frac{\pi}{4} \text{ both hold.}$$

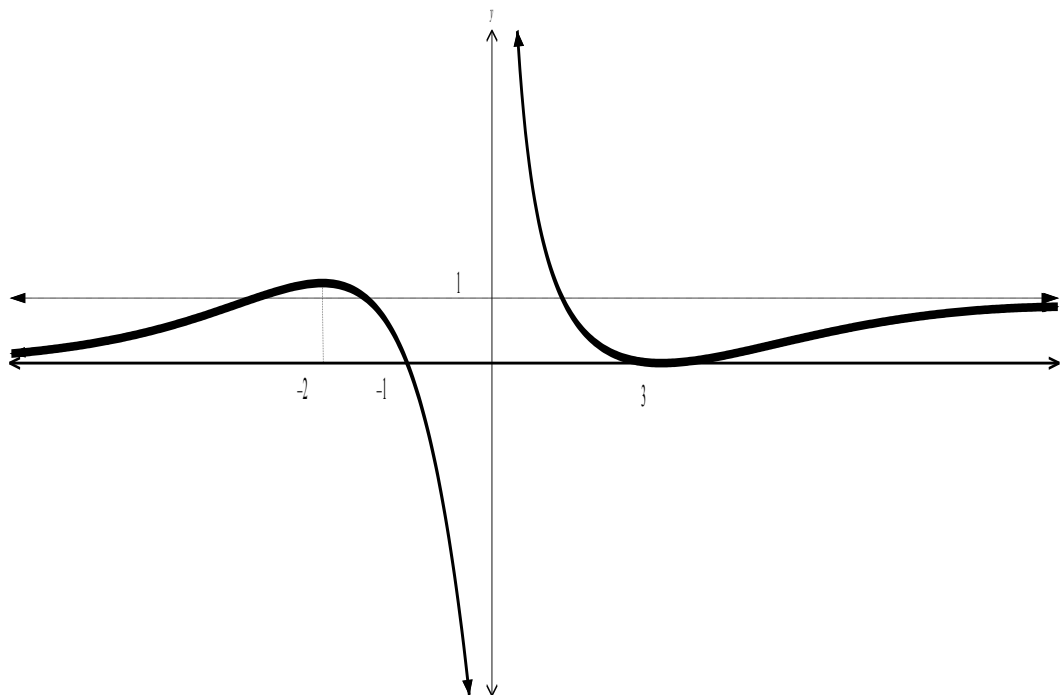
End of Question 2

Question 3 (15 Marks)

Begin a NEW booklet

Marks

a)



The diagram shows the graph of the function $y = f(x)$ which has asymptotes, vertically at $x = 0$ and horizontally at $y = 1$ for $x \geq 0$ and at $y = 0$ for $x \leq 0$.

Draw separate sketches of the following showing any critical features.

i. $y = \frac{1}{f(x)}$ **2**

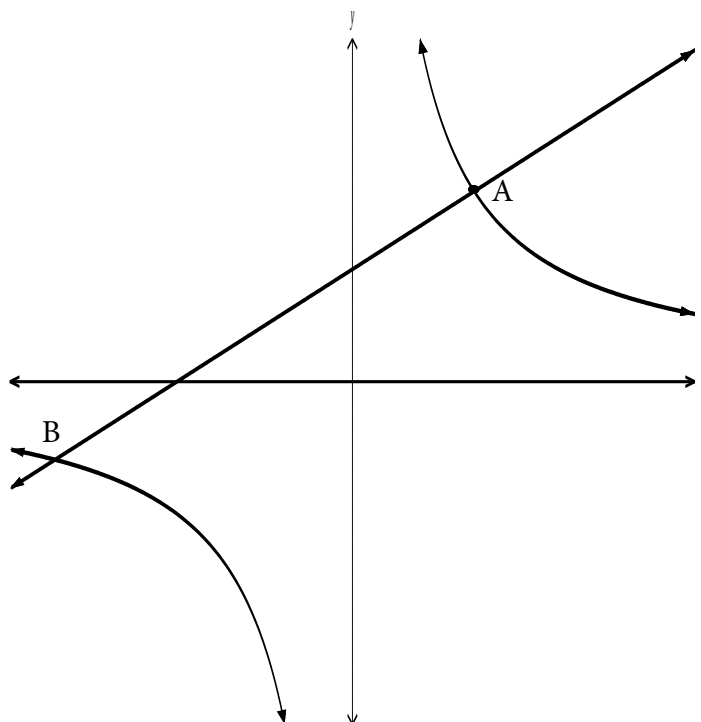
ii. $y = [f(x)]^2$ **2**

iii. $y = f'(x)$ **2**

Question 3 continues

Question 3 continued**Marks**

b)



The point $A \left(ca, \frac{c}{a} \right)$, where $a \neq \pm 1$ lies on the hyperbola $xy = c^2$. The normal through A meets the other branch of the curve at B.

- i. Show that the equation of the normal through A is

2

$$y = a^2 x + \frac{c}{a}(1 - a^4)$$

- ii. Hence if B has coordinates $\left(cb, \frac{c}{b} \right)$, show that $b = \frac{-1}{a^3}$.

3

- iii. If this hyperbola is rotated clockwise through 45° , show that the equation becomes

4

$$x^2 - y^2 = 2c^2.$$

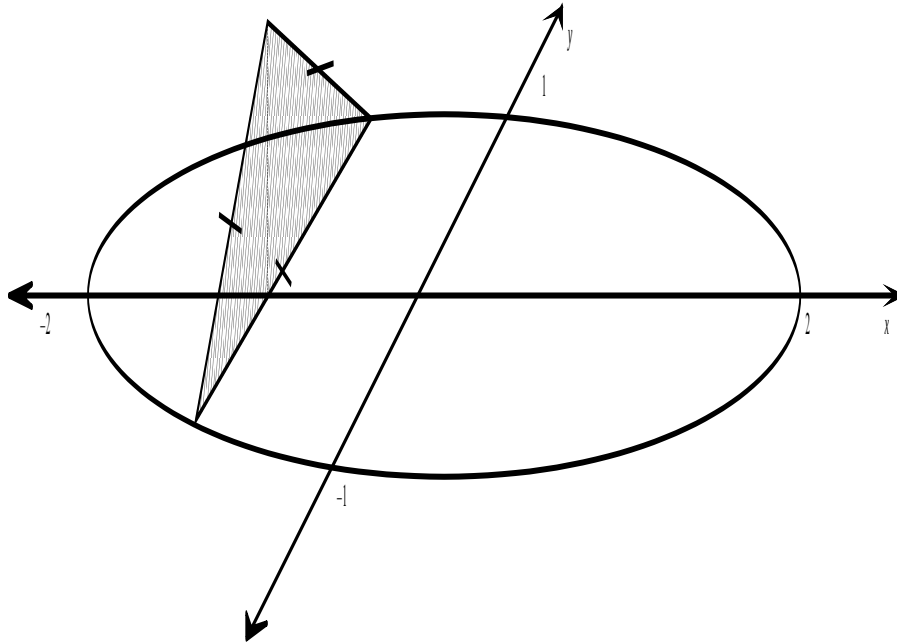
End of Question 3

Question 4 (15 Marks)

Begin a NEW booklet

Marks

- a) A solid shape has as its base an ellipse in the XY plane as shown below. Sections taken perpendicular to the X -axis are equilateral triangles. The major and minor axes of the ellipse are 4 metres and 2 metres respectively.



- i. Write down the equation of the ellipse. **1**
- ii. Show that the area of the cross-section at $x = k$ is given by **2**

$$A = \frac{\sqrt{3}}{4}(4 - k^2)$$

- iii. By using the technique of slicing, find the volume of the solid. **2**

- b) The region enclosed by the curve $y = 5x - x^2$, the x axis and the lines $x = 1$ and $x = 3$ is rotated about the y axis. By using the method of cylindrical shells, find the volume of the solid so produced. **4**

Question 4 continues

Question 4 continued**Marks**

- c) The roots of the equation $x^3 - 3x^2 + 9 = 0$ are α , β and γ .
- i. Determine the polynomial equation with roots α^2 , β^2 and γ^2 . **1**
- ii. Find the value of $\alpha^2 + \beta^2 + \gamma^2$ and hence evaluate $\alpha^3 + \beta^3 + \gamma^3$. **3**
- d) Given that the polynomial $P(x)$ has a double root at $x = \alpha$, show that the polynomial $P'(x)$ will have a single root at $x = \alpha$. **2**

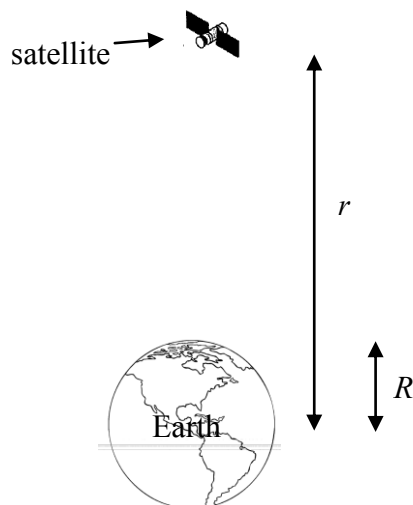
End of Question 4

Question 5 (15 Marks)

Begin a NEW booklet

Marks

a)



The gravitational force between two objects of masses m and M placed at a distance r metres apart is proportional to their masses and inversely proportional to the square of their distance apart, ie $F = k \frac{mM}{r^2}$, $k > 0$.

A satellite is to be placed in orbit so that it will rotate about the earth once every 36 hours.

i. Show that $F = \frac{R^2 mg}{r^2}$ **2**

Taking $g = 9.8 \text{ m/s}^2$ and the earth's radius $R = 6400 \text{ km}$, find:

ii. The height of the satellite from the earth's surface **3**

iii. The linear velocity of the satellite. **1**

b) By taking logarithms of both sides and then differentiating implicitly, verify the rule for differentiating the quotient $y = \frac{u(x)}{v(x)}$ is given by **2**

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Question 5 continues

Question 5 continued**Marks**

- c) i. Show that the recurrence (reduction) formula for

4

$$I_n = \int \tan^n x dx \quad \text{is} \quad I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}.$$

- ii. Hence evaluate $\int_0^{\frac{\pi}{4}} \tan^3 x dx$

3**End of Question 5**

Question 6 (15 Marks)

Begin a NEW booklet

Marks

- a) A solid of unit mass is dropped under gravity from rest at a height of H metres. Air resistance is proportional to the speed (V) of the mass. (acceleration under gravity = g)

- i. Write the equation for the acceleration of the mass. (Use k as the constant of proportionality) **1**

- ii. Show that the velocity (V) of the solid after t seconds is given by **3**

$$V = \frac{g}{k} (1 - e^{-kt})$$

- iii. By using the fact that $\frac{d}{dx} \left(\frac{1}{2} V^2 \right) = \dot{x}$, show that **4**

$$x = \frac{g}{k^2} \left[\ln \frac{g}{g - kV} - \frac{kV}{g} \right].$$

- b) Given $z = \cos \theta + i \sin \theta$, and using De Moivre's Theorem

- i. Find an expression for $\cos 4\theta$ in terms of powers of $\cos \theta$. **3**

Hint: you may use the expansion:

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

- ii. Determine the roots of the equation $z^4 = -1$. **2**

- iii. Using the fact that $z^n + \frac{1}{z^n} = 2 \cos n\theta$, find an expression for $\cos^4 \theta$ in terms of $\cos n\theta$. **2**

End of Question 6

Question 7 (15 Marks)

Begin a NEW booklet

Marks

- a) i. Prove that $\cos(k-1)\theta - 2\cos\theta\cos k\theta = -\cos(k+1)\theta$ **1**
- ii. Hence, using mathematical induction, prove that if n is a positive integer then **4**
- $$1 + \cos\theta + \cos 2\theta + \dots + \cos(n-1)\theta = \frac{1 - \cos\theta - \cos n\theta + \cos(n-1)\theta}{2 - 2\cos\theta}$$
- b) A mass of 20kg hangs from the end of a rope and is hauled up vertically from rest by winding up the rope. The pulling force on the rope starts at 250N and decreases uniformly at a rate of 10N for every metre wound up. **3**
- Find the velocity of the mass when 10 metres have been wound up.
- (Neglect the weight of the rope and take $g = 10\text{ms}^{-2}$)
- c) When a polynomial $P(x)$ is divided by $(x-1)$ the remainder is 3 and when divided by $(x-2)$ the remainder is 5. Find the remainder when the polynomial is divided by $(x-1)(x-2)$. **3**
- d) How many different words can be formed from the letters A, A, B, C, D, E, E if the word must contain: **1**
- i. All seven letters (1 mark)
- ii. Exactly four letters **3**

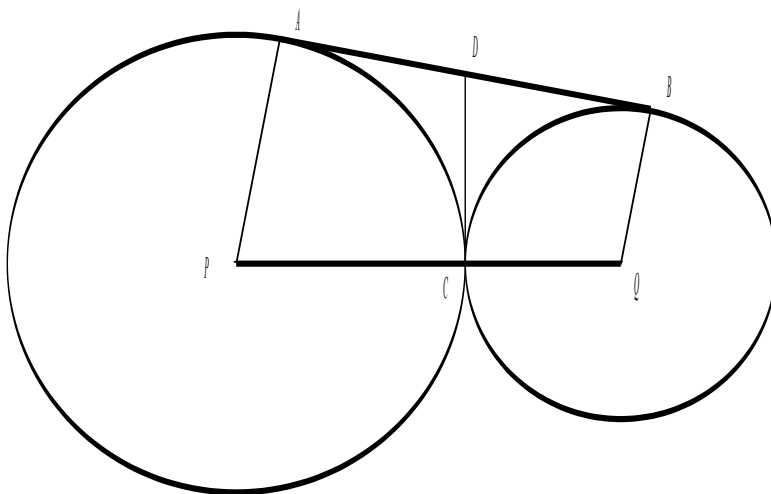
End of Question 7

Question 8 (15 Marks)

Begin a NEW booklet

Marks

a)



In the diagram PCQ is a straight line joining the centres of the circles P and Q. AB and DC are common tangents.

- i. Explain why PADC and CDBQ are cyclic quadrilaterals. **2**
- ii. Show that $\triangle ADC \parallel \triangle BQC$. **2**
- iii. Show that $PD \parallel CB$. **2**

b) Given $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$

If $P = 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta$

- i. Prove that $P \sin \frac{\theta}{2} = \sin \frac{7\theta}{2}$. **3**
- ii. Hence show that if $\theta = \frac{2\pi}{7}$ then **2**

$$P = 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta = 0$$

- iii. By writing P in terms of $\cos \theta$, prove that $\cos \frac{2\pi}{7}$ is a root of the Polynomial equation **4**

$$8x^3 + 4x^2 - 4x - 1 = 0$$

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



International
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SOLUTIONS

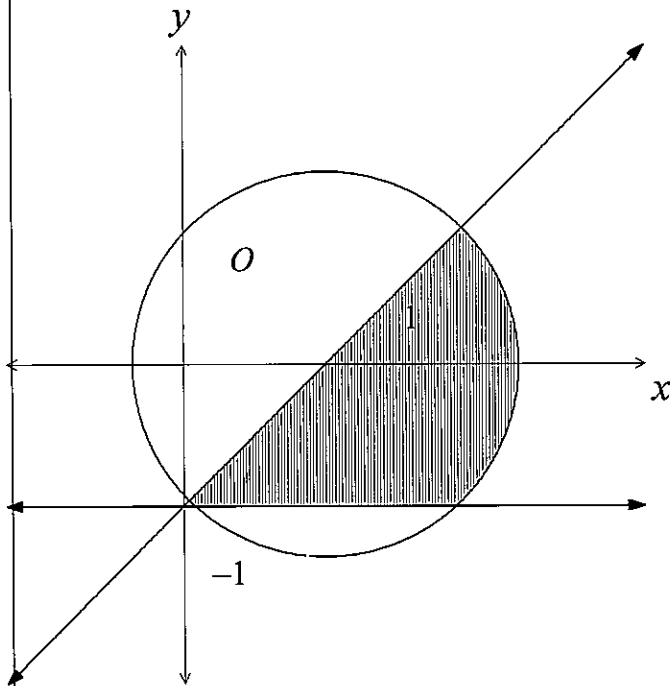
Question 1	Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment
a)	$\int_{\frac{\pi}{16}}^{\frac{\pi}{12}} \sec 4x \cdot \tan 4x \, dx = \left[\frac{\sec 4x}{4} \right]_{\frac{\pi}{16}}^{\frac{\pi}{12}}$ $= \frac{1}{4} \left[\sec \frac{\pi}{3} - \sec \frac{\pi}{4} \right]$ $= \frac{1}{4} [2 - \sqrt{2}]$ $= \frac{2 - \sqrt{2}}{4}$	1 1	Use standard integral sheet
b)	$\int x \ln x \, dx = \frac{1}{2} x^2 \ln x - \int \frac{1}{2} x^2 \cdot \frac{1}{x} \, dx$ $= \frac{1}{2} x^2 \ln x - \frac{1}{2} \int x \, dx$ $= \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + c$	1 1	
c)	$\int \frac{9x^3 + 9x^2 + 5x + 4}{3x + 1} \, dx = \int \left(3x^2 + 2x + 1 + \frac{3}{3x + 1} \right) dx$ $= x^3 + x^2 + x + \ln(3x + 1) + c$	2 1	By division
d)	i. $\frac{3x^2 - 2x - 3}{(x^2 + 9)(x - 3)} = \frac{ax + b}{x^2 + 9} + \frac{c}{x - 3}$ $3x^2 - 2x - 3 = (ax + b)(x - 3) + c(x^2 + 9)$ $x = 3 \rightarrow 18 = 18c$ $\therefore c = 1$ $x = 0 \rightarrow -3 = -3b + 9$ $\therefore b = 4$ $x = 1 \rightarrow -2 = (a + 4)(-2) + 10$ $= -2a + 2$ $\therefore a = 2$ $\therefore a = 2, b = 4, c = 1$ ii. $\int \frac{3x^2 - 2x - 3}{(x^2 + 9)(x - 3)} \, dx = \int \left(\frac{2x + 4}{x^2 + 9} + \frac{1}{x - 3} \right) dx$ $= \int \left(\frac{2x}{x^2 + 9} + \frac{4}{x^2 + 9} + \frac{1}{x - 3} \right) dx$ $= \ln(x^2 + 9) + \frac{4}{3} \tan^{-1} \frac{x}{3} + \ln(x - 3) + c$ $= \ln(x^2 + 9)(x - 3) + \frac{4}{3} \tan^{-1} \frac{x}{3} + c$	1 1 1	Any fair method <

Question 1		Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment	
e)	$\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta + \cos \theta} \quad \text{if } t = \tan \frac{\theta}{2}$ $\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$ $= \frac{1}{2} \left(1 + \tan^2 \frac{\theta}{2} \right)$ $= \frac{1}{2} (1 + t^2)$ $\frac{d\theta}{dt} = \frac{2}{1 + t^2}$ $d\theta = \frac{2 dt}{1 + t^2}$ $\left. \begin{aligned} \theta = \frac{\pi}{2}, t = 1 \\ \theta = 0, t = 0 \end{aligned} \right\}$ $\int_0^1 \frac{2 dt}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} = \int_0^1 \frac{2 dt}{1 + t^2 + 2t + 1 - t^2}$ $= \int_0^1 \frac{2 dt}{2 + 2t}$ $= [\ln(1 + t)]_0^1$ $= \ln 2 - \ln 1$ $= \ln 2$	1 1 1		
		/15		

Question 2		Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment	
a)	i. $AB = (3 + 4i)(1 - i)$ $= 3 - 3i + 4i + 4$ $= 7 + i$	1		
	ii. $\frac{A}{iB} = \frac{3 + 4i}{i(1 - i)}$ $= \frac{3 + 4i}{1 + i} \times \frac{1 - i}{1 - i}$ $= \frac{3 - 3i + 4i + 4}{2}$ $= \frac{7 + i}{2} = \frac{7}{2} + \frac{1}{2}i$	1		
	iii. Let $\sqrt{A} = a + ib$ (a and b real) $\therefore A = a^2 - b^2 + 2abi$ $\therefore a^2 - b^2 = 3$, $2ab = 4$ $ab = 2$	1		
	$\therefore b = \frac{2}{a}$ $\therefore a^2 - \frac{4}{a^2} = 3$ $a^4 - 3a^2 - 4 = 0$ $(a^2 - 4)(a^2 + 1) = 0$ $\therefore a = \pm 2$ only real solution $\therefore b = \pm 1$ $\therefore \sqrt{A} = \pm(2 + i)$	1 <		

Question 2	Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment
b)	i. $w = \sqrt{3+1} = 2$ $w = 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)$ If $\text{Arg } w = \theta$ $\cos \theta = \frac{\sqrt{3}}{2}, \sin \theta = -\frac{1}{2}$ $\therefore \theta = -\frac{\pi}{6}$ $\therefore \text{Arg } w = -\frac{\pi}{6}$ ii. $w = \sqrt{3} - i = 2\left[\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right]$ $\therefore w^5 = 32\left[\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right]$ $= 32\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)$ $= -16\sqrt{3} - 16i$	1 1 1 1	
c)	i. $OB = iOA \times 2$ $= i(x + iy) \times 2$ $= -2y + 2ix$ ii. $OC = OB + OA$ $= (-2y + 2ix) + (x + iy)$ $= (x - 2y) + (2x + y)i$ iii. $BA = BO + OA$ $= -OB + OA$ $= -(-2y + 2ix) + (x + iy)$ $= (x + 2y) + (y - 2x)i$	1 1 1	0 if signs incorrect

d)

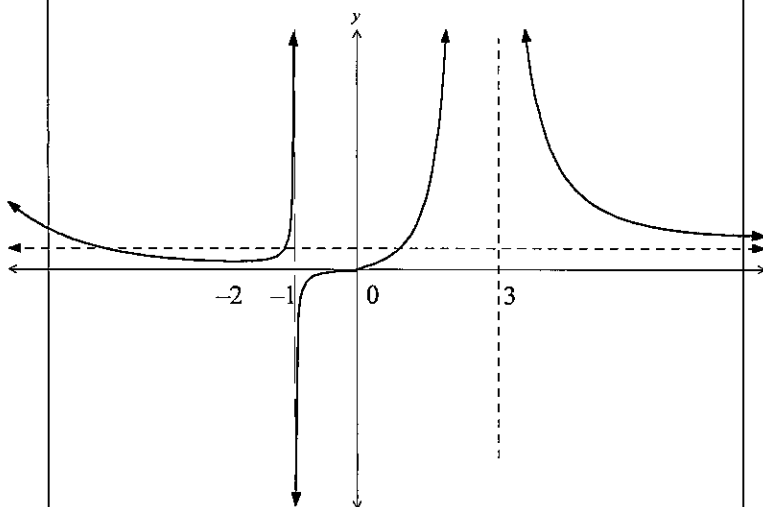
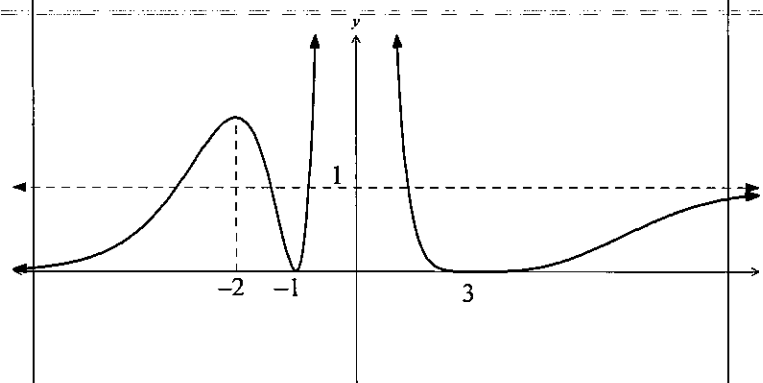
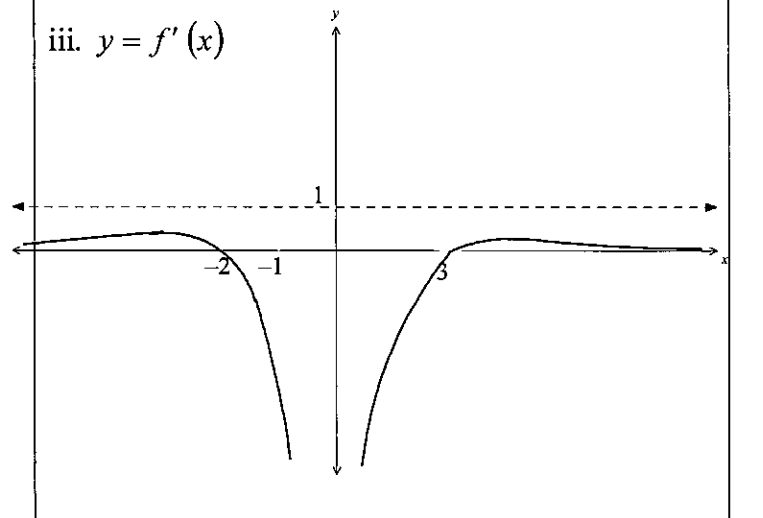


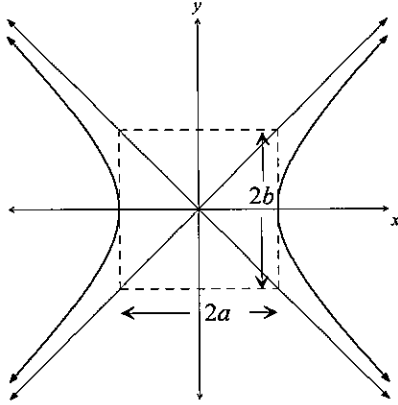
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1 for region
inside circle

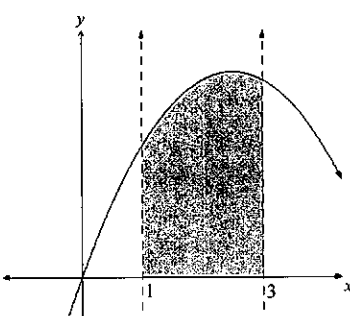
1 for between
lines

/15

Question 3	Trial HSC Examination - Mathematics Extension 2	2010	
Part	Solution	Marks	Comment
a)	i. $y = \frac{1}{f(x)}$  ii. $y = [f(x)]^2$  iii. $y = f'(x)$ 	2	2 marks each, deduct a mark for a major feature missing or incorrect, e.g. asymptotes not correct in (i)
		2	
		2	

Question 3		Trial HSC Examination - Mathematics Extension 2	2010
Part	Solution	Marks	Comment
b)	i. $y = \frac{c^2}{x}$ $\frac{dy}{dx} = -\frac{c^2}{x^2}$, at $x = ca$ $\frac{dy}{dx} = -\frac{1}{a^2}$ $\therefore$ Gradient of normal $= a^2$ $\therefore$ Equation of normal is $y - \frac{c}{a} = a^2(x - ca)$ $= a^2x - ca^3$ $y = a^2x + \frac{c}{a} - ca^3$ $y = a^2x + \frac{c}{a}(1 - a^4)$	1	
	ii. Solving $y = \frac{c^2}{x}$ with equation in (i) $\frac{c^2}{x} = a^2x + \frac{c}{a}(1 - a^4)$ $\therefore a^2x^2 + \frac{c}{a}(1 - a^4)x - c^2 = 0$	1	
	Product of roots $= -\frac{c^2}{a^2}$ The roots are $x = cb$ and $x = ca$ $\therefore c^2ab = -\frac{c^2}{a^2}$ $\therefore b = -\frac{1}{a^3}$	1	
	iii. Asymptotes become $y = \pm x$ 	1	Any logical reasoning
	Original Vertices $(\pm c, \pm c)$ Distance from O is $\sqrt{2}c$ $\therefore$ New vertices $= (\pm \sqrt{2}, 0)$ $\therefore a = \sqrt{2}c$ As it is rectangular $b = \sqrt{2}c$	1	
	$\therefore \frac{x^2}{2c^2} - \frac{y^2}{2c^2} = 1$ $\therefore x^2 - y^2 = 2c^2$	1	

Question 4	Trial HSC Examination - Mathematics Extension 2	2010
Part	Solution	Marks
a)	i. $a = 2, b = 1$ $\therefore$ Equation is $\frac{x^2}{4} + y^2 = 1$ ii. At $x = k, \quad y^2 = 1 - \frac{k^2}{4}$ $\therefore y = \pm \sqrt{\frac{4 - k^2}{4}}$ $\therefore$ Length of side of triangle = $\sqrt{4 - k^2}$ $\therefore$ Area = $\frac{1}{2} \sqrt{4 - k^2} \cdot \sqrt{4 - k^2} \sin 60^\circ$ $= \frac{1}{2} (4 - k^2) \cdot \frac{\sqrt{3}}{2}$ $= \frac{\sqrt{3}}{4} (4 - k^2)$	1 1 1
	iii. Let slice thickness = δk $\therefore$ Volume of slice $\delta V = \frac{\sqrt{3}}{4} (4 - k^2) \cdot \delta k$ $\therefore V = \int_{-2}^2 \frac{\sqrt{3}}{4} (4 - k^2) dk$ $= \frac{\sqrt{3}}{4} \left[4k - \frac{k^3}{3} \right]_{-2}^2$ $= \frac{\sqrt{3}}{4} \left[\left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) \right]$ $= \frac{\sqrt{3}}{4} \cdot \frac{32}{3}$ Volume = $\frac{8\sqrt{3}}{3}$ units³	1 1

Question 4		Trial HSC Examination - Mathematics Extension 2	2010
Part	Solution	Marks	Comment
b)	 At $x = k$ $y = 5k - k^2$ Let thickness of shell be δk $\therefore$ Volume of shell $\delta V = \pi (k^2 - (k - \delta k)^2) y$ $= \pi (2k\delta k - \delta k^2) y$ As $\delta k \rightarrow 0$ $V = \int_1^3 2\pi \cdot k \cdot y \, dk$ $= 2\pi \int_1^3 k (5k - k^2) \, dk$ $= 2\pi \left[\frac{5}{3} k^3 - \frac{k^4}{4} \right]_1^3$ $= 2\pi \left[\left(45 - \frac{81}{4} \right) - \left(\frac{5}{3} - \frac{1}{4} \right) \right]$ Volume = $\frac{140\pi}{3}$ units³	1 1 1 1	Can use formula

Question 4		Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment	
c)	i. $X = x^2$ $\therefore x = \sqrt{X}$ $\therefore X\sqrt{X} - 3X + 9 = 0$ $X(\sqrt{X} - 3) = -9$ $\sqrt{X} - 3 = \frac{-9}{X}$ $\sqrt{X} = \frac{-9}{X} + 3$ $X = \frac{81}{X^2} - \frac{54}{X} + 9$ $X^3 = 81 - 54X + 9X^2$ Required equation is $x^3 - 9x^2 + 54x - 81 = 0$	1	Any method	
	ii. from equation in (i) sum of roots is given by $\alpha^2 + \beta^2 + \chi^2 = \frac{-b}{a} = 9$	1	Any method	
	Now, in original equation $\left. \begin{aligned} \alpha^3 - 3\alpha^2 + 9 &= 0 \\ \beta^3 - 3\beta^2 + 9 &= 0 \\ \chi^3 - 3\chi^2 + 9 &= 0 \end{aligned} \right\} \text{ as } x = \alpha, \beta, \chi \text{ are roots}$ Adding, $\alpha^3 + \beta^3 + \chi^3 - 3(\alpha^2 + \beta^2 + \chi^2) + 27 = 0$ $\alpha^3 + \beta^3 + \chi^3 - 3(9) + 27 = 0$ $\therefore \alpha^3 + \beta^3 + \chi^3 = 0$	1		
d)	Let $P(x) = (x - \alpha)^2 Q(x)$ $\therefore P'(x) = 2(x - \alpha)Q(x) + (x - \alpha)^2 Q'(x)$ $= (x - \alpha)[2Q(x) + (x - \alpha)Q'(x)]$ $\therefore P'(x)$ has a single root at $x = \alpha$	1 1		
		/15		

5(a)

Rotate around earth every 36 hours.

$$\begin{aligned} \text{i) Period: } T &= 36 \times 3600 \\ &= 129600 \\ &= 1.296 \times 10^5 \text{ s.} \end{aligned}$$

$$T = \frac{2\pi}{\omega}$$

$$\omega = \frac{2\pi}{T} = 4.84814 \times 10^{-5}$$

$$F = k \frac{Mm}{r^2}$$

at earth's surface, $r = R$, $F = mg$ 1 mark.

$$mg = k \frac{Mm}{R^2}$$

$$kMm = R^2 mg$$

$$\text{so } F = \frac{R^2 mg}{r^2}$$

1 mark.

ii) Now F is centripetal towards centre of earth

$$\text{ie } F = m\omega^2 r$$

$$\text{so } m\omega^2 r = \frac{R^2 mg}{r^2}$$

$$r^3 = \frac{R^2 mg}{m\omega^2} = \frac{R^2 g}{\omega^2}$$

$$= \frac{9.8 \times (6.4 \times 10^6)^2}{(4.8 \times 10^{-5})^2}$$

$$= 1.74 \times 10^{23} \text{ or } 1.742 \times 10^{23}$$

$$\therefore r = 55827701.72 \text{ m}$$

$$\text{or } 55851458 \text{ m}$$

$$\approx 55828 \text{ km}$$

$$= 55851 \text{ km}$$

1 mark.

so height above earth's surface

$$= 55851 - 6400 = 49451 \text{ kms}$$

∴ satellite needs to be 49451 kms above earth's surface.

1 mark

$$\text{iii) } V = r\omega$$

$$= 55851 \times 4.84812 \times 10^{-5}$$

$$= 2.7 \text{ m/s}$$

1 mark

$$\begin{array}{r} 55828 \\ - 6400 \\ \hline 49428 \text{ km} \end{array}$$

b)	Let $y = \frac{u(x)}{v(x)}$ $\therefore \ln y = \ln [u(x)] - \ln [v(x)]$ $\frac{1}{y} \cdot \frac{dy}{dx} = \frac{u'(x)}{u(x)} - \frac{v'(x)}{v(x)}$ $\frac{v(x)}{u(x)} \cdot \frac{dy}{dx} = \frac{u'(x)}{u(x)} - \frac{v'(x)}{v(x)}$ $\frac{dy}{dx} = \frac{u'(x)}{v(x)} - \frac{v'(x) \cdot u(x)}{[v(x)]^2}$ $\therefore \frac{dy}{dx} = \frac{v(x) \cdot u'(x) - u(x) \cdot v'(x)}{[v(x)]^2}$	1	
c)	i. $I_n = \int \tan^n x \, dx$ $= \int \tan^2 x \cdot \tan^{n-2} x \, dx$ $= \int (\sec^2 x - 1) \tan^{n-2} x \, dx$ $= \int \sec^2 x \tan^{n-2} x \, dx - \int \tan^{n-2} x \, dx$ $= \frac{1}{n-1} \tan^{n-1} x - I_{n-2}$ ii. $\int_0^{\frac{\pi}{4}} \tan^3 x \, dx = \left[\frac{1}{2} \tan^2 x \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan x \, dx$ $= \frac{1}{2} [1 - 0] + [\ln (\cos x)]_0^{\frac{\pi}{4}}$ $= \frac{1}{2} + \left[\ln \left(\frac{1}{\sqrt{2}} \right) - \ln 1 \right]$ $= \frac{1}{2} + \ln \frac{1}{\sqrt{2}}$	1 1 1 1 1 1	
		/15	

Question 6		Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment	
a)	i. $\ddot{x} = g - kv$	1		
	ii. $\ddot{x} = \frac{dv}{dt} = g - kv$ $\frac{dt}{dv} = \frac{1}{g - kv}$ $t = -\frac{1}{k} \ln (g - kv) + c$ When $t = 0, v = 0$ $\therefore c = \frac{1}{k} \ln g$ $\therefore t = \frac{1}{k} \ln g - \frac{1}{k} \ln (g - kv)$ $kt = \ln \left(\frac{g}{g - kv} \right)$ $e^{kt} = \frac{g}{g - kv}$	1		
	$ge^{kt} - kve^{kt} = g$ $ge^{kt} - g = kve^{kt}$ $g(e^{kt} - 1) = kve^{kt}$ $\therefore v = \frac{g}{k} (1 - e^{-kv})$	1		

Question 6	Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment
a)	iii. $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \cdot \frac{dv}{dx}$ $= v \frac{dv}{dx}$ $\therefore v \frac{dv}{dx} = g - kv$ $\frac{dv}{dx} = \frac{g - kv}{v}$ $\frac{dx}{dv} = \frac{v}{g - kv}$ $\frac{dx}{dv} = \frac{1}{k} \left[\frac{v}{\frac{g}{k} - v} \right] = \frac{1}{k} \left[\frac{v - \frac{g}{k}}{\frac{g}{k} - v} + \frac{\frac{g}{k}}{\frac{g}{k} - v} \right]$ $\frac{dx}{dv} = \frac{1}{k} \left[-1 + \frac{\frac{g}{k}}{\frac{g}{k} - v} \right]$ $\therefore x = \frac{1}{k} \left[-v - \frac{g}{k} \ln \left(\frac{g}{k} - v \right) \right] + c$ When $x = 0, v = 0$ $\therefore c = \frac{1}{k} \cdot \frac{g}{k} \ln \frac{g}{k}$ $\therefore x = \frac{1}{k} \left[\frac{g}{k} \ln \frac{g}{k} - v - \frac{g}{k} \ln \left(\frac{g}{k} - v \right) \right]$ $= \frac{1}{k} \left[\frac{g}{k} \ln \left(\frac{\frac{g}{k}}{\frac{g}{k} - v} \right) - v \right]$ $x = \frac{g}{k^2} \left[\ln \left(\frac{g}{g - kv} \right) - \frac{kv}{g} \right]$	1 1 1 1	

Question 6		Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment	
b)	i. $z^4 = \cos 4\theta + i \sin 4\theta$ also by expansion $= c^4 + 4ic^3s - 6c^2s^2 - 4ics^3 + s^4$ Where $c = \cos \theta$ and $s = \sin \theta$ $= c^4 + s^4 - 6c^2s^2 + 4i(c^3s - cs^3)$ Equating real parts $\cos 4\theta = \cos^4 \theta + \sin^4 \theta - 6\cos^2 \theta \sin^2 \theta$ $= \cos^4 \theta + (1 - \cos^2 \theta)^2 - 6\cos^2 \theta (1 - \cos^2 \theta)$ $= \cos^4 \theta + 1 - 2\cos^2 \theta + \cos^4 \theta - 6\cos^2 \theta + 6\cos^4 \theta$ $= 8\cos^4 \theta - 8\cos^2 \theta + 1$	1		
	ii. $\cos 4\theta = -1$ (Equating real parts) $\therefore 4\theta = \pi, -\pi, 3\pi, -3\pi$ $\theta = \frac{\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, -\frac{3\pi}{4}$	1		
	$\therefore$ Roots are $e^{-i\frac{\pi}{4}}, e^{i\frac{\pi}{4}}, e^{i\frac{3\pi}{4}}, e^{-i\frac{3\pi}{4}}$	1		
	iii. $z + \frac{1}{z} = 2 \cos \theta$ $\left(z + \frac{1}{z}\right)^4 = 16 \cos^4 \theta$ LHS $= z^4 + 4z^3 \cdot \frac{1}{z} + 6z^2 \cdot \frac{1}{z^2} + 4z \cdot \frac{1}{z^3} + \frac{1}{z^4}$ $= z^4 + \frac{1}{z^4} + 4\left(z^2 + \frac{1}{z^2}\right) + 6$ $\therefore 16\cos^4 \theta = 2\cos 4\theta + 8\cos 2\theta + 6$ $\therefore \cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$	1		
		1		

Question 7		Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment	
a)	i. $\cos [(k-1)\theta] - 2 \cos \theta \cos k\theta$ $= \cos k\theta \cos \theta + \sin k\theta \sin \theta - 2 \cos \theta \cos k\theta$ $= -(\cos k\theta \cos \theta - \sin k\theta \sin \theta)$ $= -\cos (k\theta + \theta)$ $= -\cos [(k+1)\theta]$	1		
	ii. When $n = 1$ $LHS = 1 \quad RHS = \frac{1 - \cos \theta - \cos \theta + \cos 0}{2 - 2 \cos \theta}$ $= \frac{2 - 2 \cos \theta}{2 - 2 \cos \theta} = 1 = LHS$	1		
	$\therefore$ True for $n = 1$ Assume true for $n = k$. i.e. $1 + \cos \theta + \dots + \cos [(k-1)\theta] = \frac{1 - \cos \theta - \cos k\theta + \cos [(k-1)\theta]}{2 - 2 \cos \theta}$			
	When $n = k+1$			
	$1 + \cos \theta + \dots + \cos [(k-1)\theta] + \cos k\theta$ $= \frac{1 - \cos \theta - \cos k\theta + \cos [(k-1)\theta]}{2 - 2 \cos \theta} + \cos k\theta$ $= \frac{1 - \cos \theta - \cos k\theta + \cos [(k-1)\theta] + 2 \cos k\theta - 2 \cos \theta \cos k\theta}{2 - 2 \cos \theta}$ $= \frac{1 - \cos \theta + \cos k\theta - \cos [(k+1)\theta]}{2 - 2 \cos \theta}$ $= \frac{1 - \cos \theta - \cos [(k+1)\theta] + \cos [((k+1)+1)\theta]}{2 - 2 \cos \theta}$	1		
	If true for $n = 1$ then true for $n = 1 + 1 = 2$ etc $\therefore$ By induction true for all n positive integers.	1		

Question 7		Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment	
b)	Let Pulling force = F and distance pulled up be x metres. $\frac{dF}{dx} = -10$ $F = -10x + c$ When $x = 0$, $F = 250$ $\therefore F = 250 - 10x$ At the mass $20 \ddot{x} = F - 20g$ $= 250 - 10x - 200$ $= 50 - 10x$ $\therefore \ddot{x} = \frac{5}{2} - \frac{x}{2}$ $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{5}{2} - \frac{x}{2}$ $\frac{1}{2} v^2 = \frac{5x}{2} - \frac{x^2}{4} + c$ When $x = 0$, $v = 0$, $\therefore c = 0$ $\frac{1}{2} v^2 = \frac{5x}{2} - \frac{x^2}{4}$ $v^2 = 5x - \frac{x^2}{2}$ When $x = 10$, $v^2 = 0$ $\therefore$ The mass is stationary.	1 1 1		
c)	When dividing by $(x - 1)(x - 2)$ the remainder is in the form $ax + b$. $P(x) = (x - 1)(x - 2) Q(x) + ax + b$ $P(1) = a + b = 3$ $P(2) = 2a + b = 5$ $\therefore a = 2, b = 1$ $\therefore$ Remainder is $2x + 1$	1 1 1		

7(d) A A B C D E E $n=7$ $2 \times A$ $2 \times E$

i. using all 7 letters.

$$\text{number} = \frac{7!}{2! \times 2!} = 1260$$

1 mark correct answer.

ii) using 4 letters.

1) all different: ${}^5C_4 \times 4! = 120$

2) 2 A's, no E's: ${}^3C_2 \times \frac{4!}{2!} = 36$

3) 2 A's, 1 E: ${}^3C_1 \times \frac{4!}{2!} = 36$

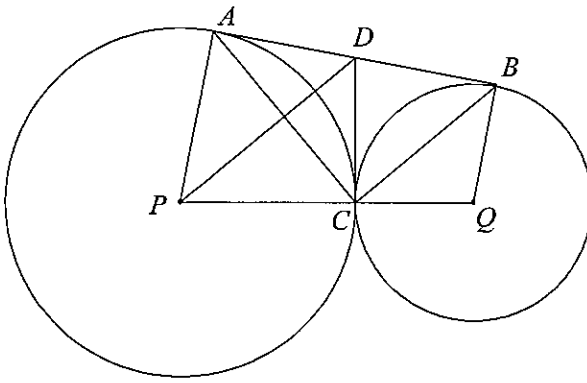
4) 2 A's, 2 E's: $\frac{4!}{2! \times 2!} = 6$

5) 2 E's, no A: ${}^3C_2 \times \frac{4!}{2!} = 36$

6) 2 E's, 1 A: ${}^3C_1 \times \frac{4!}{2!} = 36$

Total number = 270.

3 marks correct answer

Question 8	Trial HSC Examination - Mathematics Extension 2		2010
Part	Solution	Marks	Comment
a)	 i. $\angle PAD = \angle DCP = 90^\circ$ (Radius is perpendicular to tangent at point of contact) $\therefore PADC$ is cyclic (Opposite angles supplementary) Similar for $CDBQ$	2	
	ii. Let $\angle ADC = \theta$ $\therefore \angle BQC = \theta$ (Ext. angle of cyclic quadrilateral) $DA = DC$ (Equal Tangents) $\therefore \triangle ADC$ is isosceles $\therefore \angle DAC = \angle DCA = \left(90 - \frac{\theta}{2}\right)^\circ$ $BQ = CQ$ (Equal radii) $\therefore \triangle BQC$ is isosceles $\therefore \angle BCQ = \angle CBQ = \left(90 - \frac{\theta}{2}\right)^\circ$ $\therefore \triangle ADC \parallel \triangle BQC$ (AAA)	2	Any correct proof
	iii. From above $\angle APC = 180 - \theta$ (opposite $\angle$ of $PADC$) $\angle PDC = \left(90 - \frac{\theta}{2}\right)^\circ$ (PD bisects $\angle APC$) $= \angle BCQ$ (from (ii)) $\therefore PD \parallel CB$ (corresponding $\angle$ equal)	2	

Question 8		Trial HSC Examination - Mathematics Extension 2	2010
Part	Solution	Marks	Comment
b)	i. $P \sin \frac{\theta}{2}$ $= (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) \sin \frac{\theta}{2}$ $= \sin \frac{\theta}{2} + 2 \cos \theta \sin \frac{\theta}{2} + 2 \cos 2\theta \sin \frac{\theta}{2} + 2 \cos 3\theta \sin \frac{\theta}{2}$ $\therefore = \cancel{\sin \frac{\theta}{2}} + \cancel{\sin \frac{3\theta}{2}} - \cancel{\sin \frac{\theta}{2}} + \cancel{\sin \frac{5\theta}{2}} - \cancel{\sin \frac{3\theta}{2}} + \sin \frac{7\theta}{2} - \cancel{\sin \frac{5\theta}{2}}$ $= \sin \frac{7\theta}{2}$ ii. From (i) $P \sin \frac{\theta}{2} = \sin \frac{7\theta}{2}$	1 1 1	
	When $\theta = \frac{2\pi}{7}$ $P \sin \frac{\pi}{7} = \sin \pi$ $= 0$ As $\sin \frac{\pi}{7} \neq 0$ then $P = 0$ i.e. $1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta = 0$ iii. $P = 1 + 2 \cos \theta + 2(2 \cos^2 \theta - 1) + 2(4 \cos^3 \theta - 3 \cos \theta)$ $= 8 \cos^3 \theta + 4 \cos^2 \theta - 4 \cos \theta - 1$ $P = 8x^3 + 4x^2 - 4x - 1 \text{ when } x = \cos \theta$ From (ii) $P = 0 \text{ when } \theta = \frac{2\pi}{7}$ $\therefore x = \cos \frac{2\pi}{7} \text{ is a solution.}$	1 1 1 1 1	
		/15	